Zariski density of the prepared pole witnesses

Abstract

The simultaneous elliptic realization underlying the pole certificate gives more than one rational witness. Its witnesses are Zariski dense in the two retained pole parameters. Consequently finitely many additional nonzero polynomial conditions in those parameters may be imposed without strengthening the inherited arithmetic input. We prove this by independently varying odd multiples of the two realized non-torsion points. A finite-algebra corollary permits simultaneous nonvanishing of all conjugate second-root radicands whenever the pole input is already a unit in that algebra. The result does not assert that the other source guards are simultaneously units.

1 The realization input

Fix a prepared rational number b in the sense of [1], and write

$$B = 17b, \qquad V = B(B - b^2) > 0, \qquad V = \theta c^2,$$

where θ is the positive squarefree representative and $c \in \mathbb{Q}^\times$. For $a \neq 0$, put

$$E_a : \quad y^2 = x(x - a)(x + 3a).$$

The inherited simultaneous-realization input supplies a positive rational number n and points

$$Q_1 \in E_n(\mathbb{Q}), \qquad Q_2 \in E_{n\theta}(\mathbb{Q}), \qquad x(Q_1), x(Q_2) \in B\mathbb{Q}^{\times 2}, \tag{1}$$

outside the two-torsion. Its construction includes a fresh prime of odd valuation in n outside the support of θ; thus neither n nor $n\theta$ is a rational square. These are the same realized twists used in the source pole-chart lemma [2].

Lemma 1.1. *If $a \in \mathbb{Q}^\times$ is not a square, then $E_a(\mathbb{Q})_{\mathrm{tors}} = E_a[2](\mathbb{Q})$.*

Proof. There is full rational two-torsion. Mazur's torsion theorem [4] therefore leaves the groups $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/(2m)\mathbb{Z}$, with $1 \leq m \leq 4$. It suffices to exclude points of orders three and four.

The three-division polynomial is

$$3x^4 + 8ax^3 - 18a^2x^2 - 9a^4.$$

A rational root would give a rational zero of $3z^4 + 8z^3 - 18z^2 - 9$ on putting $z = x/a$. This polynomial has no root modulo 5: its five values are $1, 4, 1, 3, 3$. A rational zero cannot have negative 5-adic valuation, since the leading coefficient is a 5-adic unit. Thus no rational zero exists.

If a rational two-torsion point with abscissa e_i is a double, its differences $e_i - e_j$ and $e_i - e_k$ from the other two roots must be rational squares. This necessary condition follows directly from the doubling formula for a cubic with three rational roots. At $e_i = 0$, the differences $-a, 3a$ have opposite signs. At $e_i = -3a$, their ratio is $3/4$, not a rational square. At $e_i = a$, they are $a, 4a$, which cannot both be squares under the hypothesis. Hence no point of order four exists, and the stated torsion group follows. □

In particular, both points in (1) have infinite order. For the density theorem it is enough to assume directly the existence of the two infinite-order points in that display. This formulation makes the exact arithmetic dependency explicit.

2 Two independent infinite sets of witnesses

The squareclass map

$$\alpha_a : E_a(\mathbb{Q}) \longrightarrow \mathbb{Q}^\times/\mathbb{Q}^{\times 2}, \quad O \longmapsto 1, \quad (0,0) \longmapsto -3a^2, \quad (x,y) \longmapsto x \quad (x \neq 0)$$

is a homomorphism. This follows from the chord-and-tangent law: the product of the three abscissas cut out by a line with nonzero intercept is a square. The lines through $(0,0)$ give its displayed exceptional value; see also [3, Lemma 2.2]. Thus every odd multiple of Q_i has the same nonzero abscissa squareclass B. The abscissas of the positive odd multiples form an infinite set, since an abscissa determines at most two points and Q_i has infinite order.

Choose arbitrary positive odd multiples and write

$$Q_1' = (Bu^2, y_1) \in E_n(\mathbb{Q}), \qquad Q_2' = (Bv^2, y_2) \in E_{n\theta}(\mathbb{Q}), \qquad u, v \in \mathbb{Q}^\times.$$

Define

$$e = \frac{n}{Bu^2}, \qquad p = \frac{y_1}{Bu^3}, \qquad s = \frac{cv}{u}, \qquad q = \frac{c^2 y_2}{B^2 u^2 v}. \tag{2}$$

Rescaling the first curve by u^{-1} and the second by c/u gives the identities

$$p^2 = B(1-e)(1+3e), \qquad Bq^2 = (s^2 - eV)(s^2 + 3eV).$$

None of p, q, s vanishes, because neither multiple is torsion. Since $n, B > 0$ and the positive abscissa of a non-torsion point on E_n is greater than n, one has $0 < e < 1$. Consequently the single rational chart is available:

$$\tau = \frac{p}{e-1}, \qquad e = \frac{\tau^2 - B}{\tau^2 + 3B}, \qquad p = -\frac{4B\tau}{\tau^2 + 3B}. \tag{3}$$

Theorem 2.1. *Assume the realization input* (1)*. The set of pairs* $(\tau, s) \in \mathbb{Q}^2$ *obtained from rational pole witnesses with* $pqs \neq 0$ *is Zariski dense in* $\mathbb{A}^2$*. In particular, if* $F_1, \ldots, F_r \in \overline{\mathbb{Q}}[T, S]$ *are nonzero polynomials, there is a witness in the single chart* (3) *with*

$$F_j(\tau, s) \neq 0 \qquad (1 \leq j \leq r).$$

Proof. Varying the positive odd multiple Q_1' gives infinitely many values of u^2, and therefore infinitely many values of $e = n/(Bu^2)$. Formula (3) shows that the corresponding τ values also form an infinite set.

Fix one such choice of Q_1' and hence one value of τ. Independently varying the positive odd multiple Q_2' gives infinitely many values of v^2, and hence infinitely many values of $s = cv/u$.

Suppose a polynomial $F(T, S)$ vanishes on every pair so obtained. For each of the infinitely many fixed values τ, the polynomial $F(\tau, S)$ has infinitely many zeros and is therefore zero. Its coefficient polynomials in T vanish at infinitely many values, so each is zero. Thus $F = 0$, proving density. Applying this argument to the nonzero product $F_1 \cdots F_r$ proves the last assertion, equally over any algebraic extension of $\mathbb{Q}$. □

Corollary 2.2. *For any finite list of independently prepared pole inputs, the product of their rational witness sets is Zariski dense in the product of their parameter planes. Any finite collection of nonzero polynomial conditions involving all those parameters can therefore be avoided simultaneously.*

Proof. Products of Zariski-dense subsets are Zariski dense, by applying the one-factor coefficient argument successively. □

3 Nonvanishing in a finite algebra

Let A be a finite-dimensional commutative $\mathbb{Q}$-algebra of dimension $d > 0$. Reducedness is not assumed. Suppose $\mathcal{B}$ is a unit of A, and let $\mathcal{V} \in A$. For indeterminates T, S, put

$$\mathcal{D}(T) = T^2 + 3\mathcal{B}, \qquad ⌉(T) = \frac{T^2 - \mathcal{B}}{T^2 + 3\mathcal{B}}$$

in the localization where $\mathcal{D}$ is a unit, and define

$$\mathcal{R}(T, S) = \frac{(S^2 - ⌉(T)\mathcal{V})(S^2 + 3⌉(T)\mathcal{V})}{\mathcal{B}}.$$

The algebra norm is the determinant of multiplication on A.

Proposition 3.1. *The polynomial* $\mathrm{Norm}_A(\mathcal{D}(T))$ *is monic of degree* $2d$*. For every rational* t *where this norm is nonzero,*

$$\mathrm{Norm}_A(\mathcal{R}(t, S)) \in \mathbb{Q}[S]$$

has degree $4d$ *and nonzero leading coefficient* $\mathrm{Norm}_A(\mathcal{B}^{-1})$*.*

Proof. The multiplication matrix of $\mathcal{D}(T)$ is $T^2 I + 3M_{\mathcal{B}}$, giving the first assertion. At an admissible t, the radicand $\mathcal{R}(t, S)$ has degree four in S and leading coefficient $\mathcal{B}^{-1}$. Its multiplication matrix therefore has leading coefficient $M_{\mathcal{B}^{-1}}$, whose determinant is nonzero. The determinant has degree $4d$ with the asserted leading term. □

Suppose a rational character $A \to \mathbb{Q}$ sends $\mathcal{B}$ to the actual B of a prepared pole input, and sends $\mathcal{V}$ to its actual V. Theorem 2.1 and Proposition 3.1 then allow an actual rational pole witness with both displayed norms nonzero. Hence its chart denominator and its second-root radicand are units simultaneously at every geometric character of A. This remains true for several poles by Corollary 2.2.

The hypothesis that $\mathcal{B}$ is already a unit is essential to this application. The density theorem proves avoidance of nonzero specialized polynomial conditions; it cannot remove a condition that vanishes identically after the other source coordinates have been fixed. In particular, it does not establish simultaneous nonvanishing of all point-chart discriminants or all source denominators. Those require a separate argument.

References

[1] *Arithmetic dependencies of the construction with 29 variables*, project paper, version 8, 8 November 2026.

[2] *A defining polynomial in twenty-seven variables: an existence theorem with a uniform degree bound*, project paper, version 12, 9 November 2026, pole-chart lemma.

[3] *No constant pole parameter is complete*, project paper, version 14, 9 November 2026.

[4] B. Mazur, *Modular curves and the Eisenstein ideal*, Publications Mathématiques de l'IHÉS **47** (1977), 33–186, `https://doi.org/10.1007/BF02684339`.